PUNJAB
INTEGRATED RISK MONITORING AND EARLY WARNING SYSTEM

Sector: Agriculture & Social Protection
Hazard adressed: Flood
Targeted assets: Livestock, all population groups and households

Description: This measure establishes an integrated flood monitoring and early warning system across 8 Punjab districts through three components: Discharge Measuring Systems (DMS) at 14 major river bridges; Automated Weather Stations (AWS); and community dissemination infrastructure (sirens, SMS gateways, staff training) for 22.4M people. Data centres are excluded from scope, assumed covered under existing government initiatives. The system enables timely evacuation and protective action, significantly reducing flood-related casualties and asset losses.

Priority Districts: Dera Ghazi Khan, Mianwali, Rajanpur, Rahim Yar Khan, Chiniot, Toba Tek Singh, Jhang, Layyah.

TECHNICAL SPECIFICATIONS/DIMENSION

Unit of implementation: Per district.

Scale assumptions: Entire targeted districts.

Key design parameters: IInstallation of DMS on major bridges and AWS in districts vulnerable to hill torrents, integration with existing DMS already installed at major barrages and dams both within the Punjab province as well as those installed on key locations in KP & GB provinces. Data centres not included in the cost estimates because data centres being/ already established under other initiatives to be leveraged. Early warnings dissemination infrastructure for the target districts included.

Time horizon: 2026–2050.

IMPACT

Risk reduction level: High Effectiveness – significant reduction.

Time to effectiveness: System becomes operational once sensors, AWS, and communication infrastructure are installed and staff trained. Effectiveness is contingent on community awareness and emergency response capacity.

Expected impact: 95% reduction in flood damage for all population groups through timely warning and evacuation; applicable to 75% of at-risk population (PAA = 0.25); no direct flood intensity or frequency reduction.

BENEFICIARIES

Primary beneficiaries: People in 8 target districts; agricultural communities with advance warning of flood events; district disaster management authorities.

Distributional aspects: Particularly beneficial for rural and remote communities with limited access to formal early warning channels; women, children, elderly and PWDs who may be less mobile and more at risk from sudden flooding.

COST ASSUMPTIONS

Unit cost: USD 170,830 per district but involves leveraging the instrumentation in upstream districts.

Capital Expenditure (CAPEX):
USD 0.14 million per year (2026 – 2035).

Operational Expenditure (OPEX):
USD 0.16 million per year (2031 – 2050).

CO-BENEFITS & TRADE-OFFS

Environmental: Real-time hydrological data supports better water resource management and environmental monitoring beyond early flood warning.

Social: Significant life-safety benefits through timely evacuation; reduced trauma from unexpected flood events; improved community preparedness and flood risk awareness; particularly beneficial for vulnerable groups (elderly, women, children, people with disabilities).

Economic: Avoided losses from timely evacuation of livestock and household assets; reduced emergency response costs with better advance warning; improved agricultural decision-making from real-time weather data.

Potential risks: False alarms reduce community trust over time; effectiveness depends on last-mile communication reliability; risk of data centre dependency not covered by this measure.

IMPLEMENTATION CONSIDERATIONS

Enabling conditions: Reliable communication infrastructure (mobile coverage); community training and drills; inter-agency data sharing protocols between meteorology, irrigation, and PDMA; integration with 23 existing barrages DMSs.

Required institutions: PDMA Punjab (lead); Pakistan Meteorological Department (PMD); Punjab Irrigation Department; Pakistan Telecommunications Authority (for SMS gateway); District Disaster Management Authorities (DDMAs).

Barriers: Communication dead zones in rural areas; maintenance of remote sensors in flood-prone areas; community compliance with warnings; coordination between multiple agencies; data centre dependency not covered by this measure's CAPEX.

Note: This factsheet is to be used as a rapid planning and appraisal tool. All costs and impacts should be validated through field assessments, stakeholder consultation and detailed feasibility studies.